

Braillix - Master Bill of Materials

Supersedes docs/SHOPPING_LIST.md (that file still lists 4mm pen springs and 2mm bearing balls - both wrong since v7.1). Buy from this file only. Scope: **1 Brain Pod + 1 Cell**, enough to make one dot go up and come back down. Budget ₹15,000 - already spent ₹935 (J.B. Enterprises). Fastener quantities are **derived from the .scad sources and verified**, not estimated - each row names the feature it serves. Prices are realistic Indian estimates (Amazon.in / robu.in / local market), marked ~ where not quoted from a specific listing.

Status legend

	Meaning
[OK] HAVE IT	Already owned - from the ₹935 bill or stated in hand
[X] NEED TO BUY	Not owned. Where + estimated cost given
[!] HAVE BUT WRONG	Owned, but the wrong spec or not enough

1. Electronics

Component	Status	Action Needed
ESP32 DevKit V1 (30-pin, USB-C)	[OK] HAVE IT	-
5V/3A power adapter	[OK] HAVE IT	-
Barrel jack (yellow screw-terminal)	[!] HAVE BUT UNVERIFIED	[!!!] Polarity has NEVER been checked. Verify with a multimeter BEFORE powering anything. See ASSEMBLY_BIBLE Step 1. Wrong polarity kills the ESP32 instantly.
28BYJ-48 stepper motor (5V)	[OK] HAVE IT	[!] Never measured with calipers. 8 measurements needed before the cam/base_plate can be trusted - see CAD_FIT_CHECK.
ULN2003 driver module	[OK] HAVE IT	Fits the 16mm pocket only with wires soldered flat (~12mm). With vertical Dupont headers it is ~20mm and will NOT fit.
Hall sensor - MH-Sensor-Series (blue module)	[!] HAVE BUT WRONG FORM FACTOR	The base_plate pocket is 5.3 x 4.3 x 3mm , sized for a bare TO-92 (SS49E/A3144). The blue module PCB is far larger and will not fit. Fix: desolder the TO-92 off the module (free), or buy 3x bare SS49E (~₹90). Must use the analog AO output - firmware does <code>analogRead()</code> with 500/3500 thresholds, so a digital-only board breaks homing.
Neodymium magnets 8x1mm (docking)	[OK] HAVE IT	10 purchased. 4 needed per cell (2 per +/-X face).
Homing magnet 3x2mm (for the cam)	[X] NEED TO BUY	1 per cell. An 8x1mm will not fit the cam's 3.2mm pocket. Search "3x2mm neodymium magnet" - ~₹120 for 10-20 pcs.
Tactile switches 6x6x5mm	[OK] HAVE IT	3 needed (pod nav buttons). Not required for a Tier-0 demo.
Female header strips 1x15	[X] NEED TO BUY	2 for the pod DevKit mount. ~₹40. Not needed for a breadboard demo.
USB-C cable (data, not charge-only)	[X] NEED TO BUY / verify	Must carry data . Many charge-only cables look identical and silently fail to flash. ~₹150.
Resistors / capacitors	[OK] NOT NEEDED	4.7k I2C pull-ups are only for the multi-cell I2C bus, which is not in the demo. No protection parts needed for a single cell.
Pogo pin connectors	?? DEFERRED	Spec: 4-pin, 2.54mm pitch, spring-loaded, ~0.5A. Not needed for a single-cell demo. Pocket dims in the CAD are placeholders pending a real part.

2. Mechanical & fasteners

Counts derived from the SCAD sources and verified against them.

Component	Qty	Status	Serves / Action
M2.5 x 25mm bolts	4	[X] NEED TO BUY	top_plate counterbores -> base_plate standoffs -> outer_box corner bosses. Verified: 25mm gives ~9.5mm engagement. ~?60
M4 x 10mm bolts + nuts	2	[X] NEED TO BUY	Motor mounting ears (base_plate, dia 4.3 holes). ~?30
M2 x 8mm self-tap	2	[X] NEED TO BUY	Pod lid -> shell bosses. [!!] NOT M2x6 - through a 4mm lid that leaves only 2mm of thread; 8mm gives 4mm. Verified. ~?40
M2 x 6mm self-tap	4	?? DEFERRED	Muscle-board bosses. Not used in the prototype (ULN2003 sits on the pocket floor).
2mm OD micro compression springs	6 + spares	[X] NEED TO BUY	[!!] CRITICAL, no substitute. 2.0mm OD, ~0.3mm stainless wire, ~4mm free length. Pen springs (4mm) cannot work - braille rows are 2.6mm apart. Buy an assortment kit (200-400pcs) so you're not betting on one guess. ~?500

3. Consumables

Component	Status	Action Needed
Solder wire	[X] NEED TO BUY	0.8mm, 60/40 rosin-core. Not plumbing solder. ~?150
Flux paste	[X] NEED TO BUY	Not strictly required, but makes soldering to ULN2003 pins far easier for a beginner. ~?100
Superglue (CA) or 5-min epoxy	[X] NEED TO BUY	Glues the resin dot_insert into the PETG top plate, and the magnets. Epoxy recommended - it gives you time to seat the tile square. ~?80
Heat-shrink tubing (1-3mm assorted)	[X] NEED TO BUY	Insulating soldered joints. ~?80
Electrical tape	[X] NEED TO BUY	~?30
Silicone grease	[X] NEED TO BUY	Cam tracks, to reduce stall risk. A tiny amount. ~?80
Dupont jumper wires	[OK] HAVE IT	2m wire + spares purchased. For counts see below.
Breadboard	[X] NEED TO BUY	Yes, still needed - the Tier-0 demo is breadboard-based, and it is the safest way to bring up wiring before anything is soldered. Half+ size. ~?100

Dupont counts for the demo wiring (13 connections total): **M-F x8** (ESP32 header -> breadboard/module), **M-M x5** (rail to rail, module to rail). F-F not needed. If your 2m wire is bare, you need crimps or you solder directly.

4. Tools

Tool	Status	Action Needed
Soldering iron	[OK] HAVE IT	You stated basic experience and an iron. 25-60W with a conical/chisel tip is right.

Tool	Status	Action Needed
Digital multimeter	[X] NEED TO BUY - MANDATORY	[!!] The single most important purchase. Barrel-jack polarity is unverified and getting it wrong destroys the ESP32 on first power-up. Also your only way to debug a dead rail. ~?500
Digital calipers	[X] NEED TO BUY - HIGH PRIORITY	Both open project blockers are *measurements* (motor x8, barrel jack body). Without these the cam and base_plate cannot be trusted. ~?600
Wire stripper / cutter	[X] NEED TO BUY	~?200
Desoldering pump or wick	[X] OPTIONAL	Useful if you need to lift the TO-92 off the hall module, or fix a bridge. ~?120
Tweezers	[X] OPTIONAL	Strongly recommended for 2mm springs and 1mm linkages. ~?80
Small drill bits (1.5/1.7/2.5mm)	[X] OPTIONAL	FDM holes print 0.2-0.3mm undersize; these clean them out. ~?150

Soldering difficulty - the honest read

Nothing in the demo build requires fine-pitch work. The hardest joint is soldering wires flat onto the ULN2003's 2.54mm header pads - comfortably within basic skill.

The **only** fine-pitch work in the whole project is the custom muscle board (TQFP-32 at 0.8mm pitch, SOIC-16, 0402 passives). That is **beyond basic soldering** - order it assembled (PCBA) or stay on the ULN2003 module. It is not needed for the demo either way.

5. What to actually spend

Path	Contents	Cost
Cheapest working demo	multimeter, breadboard, solder wire, strippers, USB-C cable	~?1,100
+ mechanism	above + springs, homing magnet, fasteners, glue, calipers	~?2,900
Everything in this BOM	all of the above + optional tools + consumables	~?3,900

Project total if you buy everything: ?935 + ~?3,900 = ~?4,835 of ?15,000. Comfortable.

Deliberately excluded: the resin print order. Do the cheap PETG proving print and make one dot move *before* spending on resin. Every downstream decision rests on a mechanism that has never been physically tested.

6. Buy these first (this week)

- 1 **Multimeter** (~?500) - unblocks safe power-up. Nothing else can happen safely without it.
- 2 **Breadboard + USB-C data cable + solder wire** (~?400) - everything needed for the Tier-0 demo.
- 3 **Calipers** (~?600) - unblocks the motor measurements, which unblock the cam and base plate.

That is ~?1,500 and it gets you a demonstrable motor + hall + live dashboard, plus the measurements that unblock everything else.

7. Unverified - measure before buying or printing

#	What	Feeds
1	28BYJ-48: body dia, can height, shaft->body-centre offset, shaft dia, flat-to-flat, shaft length, mount-hole spacing, mount-hole dia	base_plate, mid_plate, cam bore
2	Barrel jack body WxLxH + barrel bore	pod lid jack cradle (currently PLACEHOLDER)

#	What	Feeds
3	Hall module PCB dimensions (or switch to bare TO-92)	base_plate hall pocket
4	Actual magnet diameter/thickness from the bill	dock magnet pockets (CAD assumes 8x1mm)
5	ESP32 board length/width + pin-row pitch	pod header channels
6	Pogo connector, once chosen	outer_box pogo carrier pocket